

Customer Churn Problem

Learning Objective

- Understanding Customer Churn
- Detrimental effect of churn on business health
- Measuring Customer Churn
- Strategies for Controlling Customer Churn

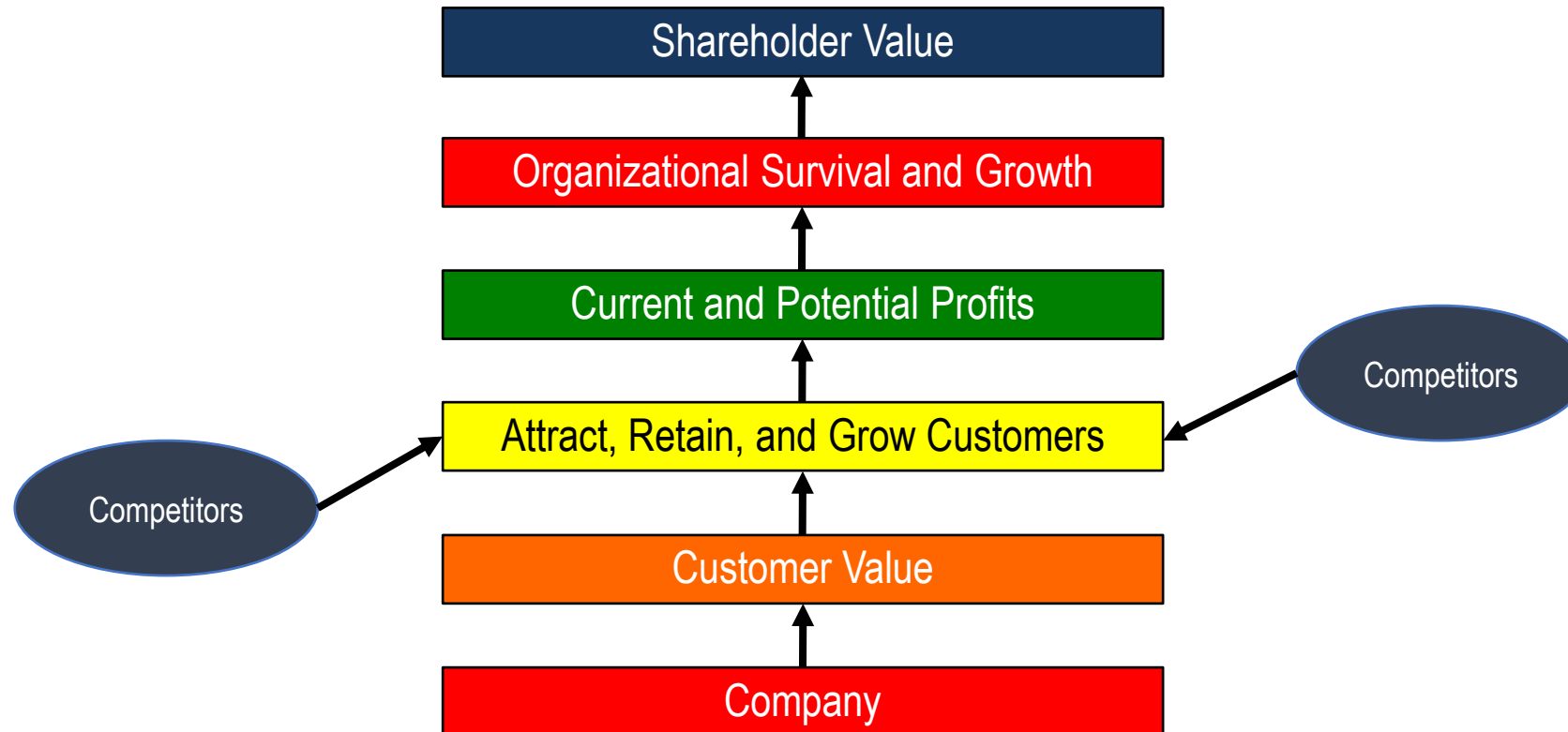
What is the purpose of business

What is the purpose of business

Drucker stated, “If we want to know what a business is, we have to start with its purpose”.

There is only one valid definition of business purpose:
to create a customer.

The Fundamental Business Model



The changing view

Old Way	New Way
Accept that some customers are difficult to address	Strive to give consistent good customer experience
Acquiring customer is critical	Retaining & growing profitable customer is critical
Firm manages products	Firm manages customers
Firm should attract, retain & grow all customers	Firm should fire some customer and be selective
Measure Product profitability	Measure customer profitability
Plant and equipment are core assets	Customers are firm's core assets
Product are at the heart of customer decision making	Customer are at the heart of customer decision making
Product profitability drive incentive system	Customer profitability drive incentive system
Zero or negative rewards for loyalty	Loyalty incentives very common

The Value of Customers

- **Customers are the firm's core assets**

When the firm creates value for customers, and retains them ...

Customers create value for the firm

- **Customers are the critical source of cash inflows**

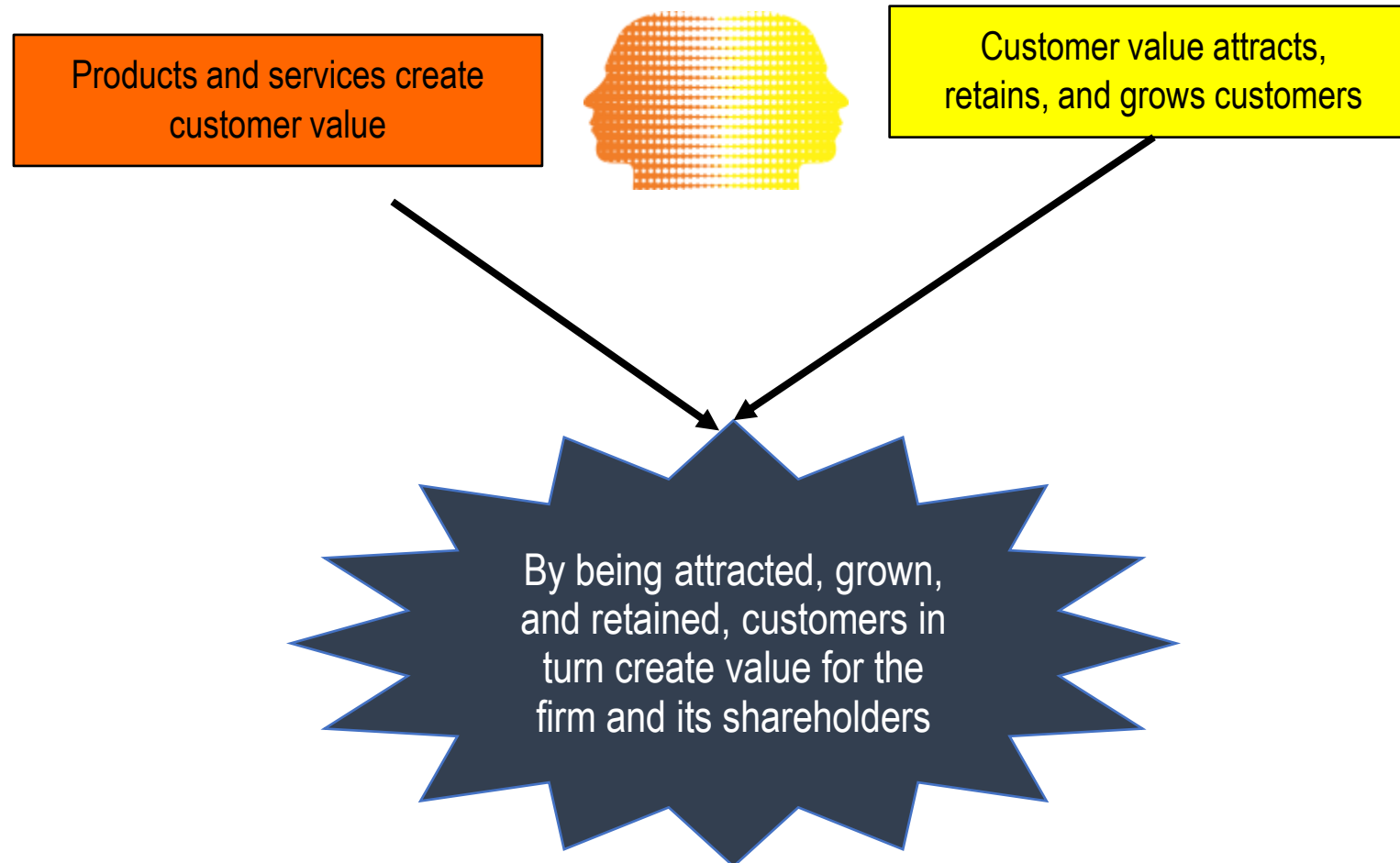
They buy more products

So long as the firm continues to deliver customer value

- **Globalization is driving increased emphasis on creating shareholder value**

The firm must determine how to use CLV to enhance shareholder value

Two Faces of Value



Customer Lifetime Value (CLV)

Customer lifetime value depends on three factors:

- Profit margin
- Retention rate
- Discount rate

$$CLV = \frac{r \times m}{(1 + d - r)}$$

$$m_1 = m_2 = m_3 \dots = m$$

$$r_1 = r_2 = r_3 \dots = r$$

Customer Lifetime Value (CLV)

$$\text{CLV} = \frac{r \times m}{(1 + d - r)}$$

Profit margin

$$\text{Margin multiple} = \times \frac{r}{(1 + d - r)}$$

$$\text{CLV} = \frac{m \times r}{(1 + d - r)}$$

Increase profit margin, m

Increase customer retention rate, r

Reduce discount rate, d

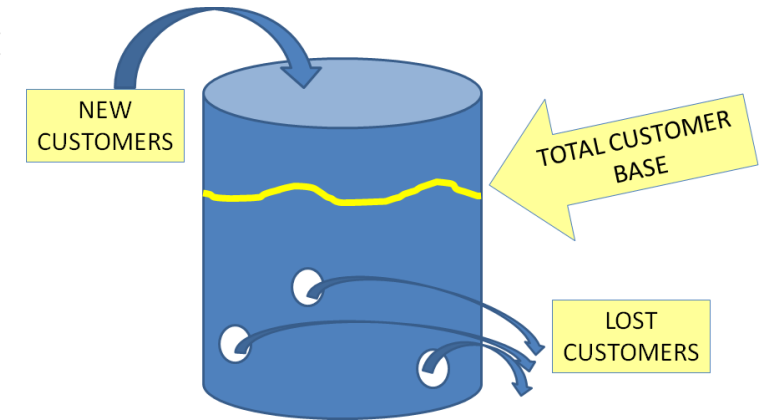
Customer Lifetime Value (CLV)

Margin multiple values

Discount Rate (d)				
Retention Rate (r)	8%	12%	16%	20%
60%	1.25	1.15	1.07	1.00
70%	1.84	1.67	1.52	1.40
80%	2.86	2.50	2.22	2.00
90%	5.00	4.09	3.46	3.00
95%	7.31	5.59	4.52	3.80

The Leaky Bucket Theory

- The leaky bucket theory is more of an analogy than a theory.
- Dynamic situation –constant flow of new customers and a consistent loss of existing customers.
- Two approaches can be taken to maximize the amount of water in the bucket (that is, the size of the customer base).
 - Increasing the water flow into the bucket – increasing the number of new customers acquired.
 - Reduce or plug some of the holes in the bucket to reduce the water loss – concentrating on customer retention.



The leaky bucket theory is a good analogy for the marketing challenge that firms face:

Deciding the proportion of their marketing expenditure and activities that should be focused upon acquiring new customers versus retaining existing customers.

Why Are Customers So Important for the Firm?

Business model

The Business Model Canvas

Designed for:		Designed by:		Date:	Version:
Key Partnerships 	Key Activities 	Value Propositions 	Customer Relationships 	Customer Segments 	
	Key Resources 		Channels 		
Cost Structure 			Revenue Streams 		



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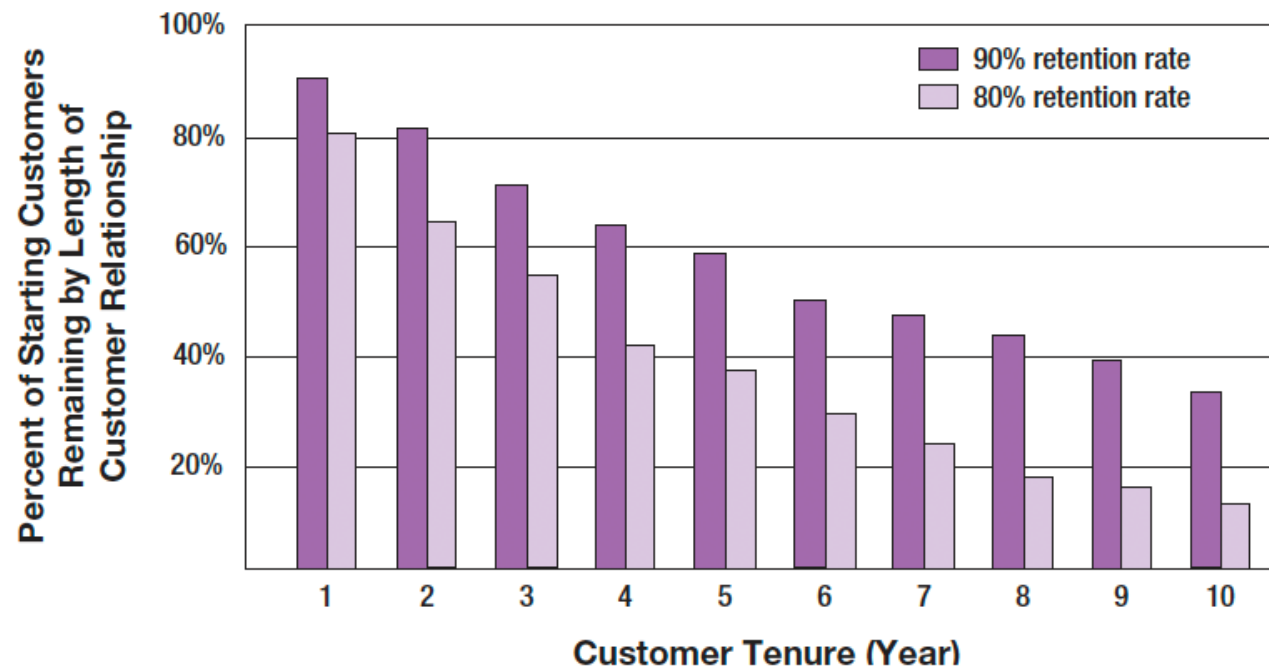
Customer Retention

Small changes in customer retention drive major changes in sales and market share

- **CLV: *Increase customer retention by 5 percent***
 - Margin multiple values – reduce discount rate/increase retention rate
 - Impact on firm growth
 - Impact of market share
 - Duration effect
 - Relationship to customer satisfaction

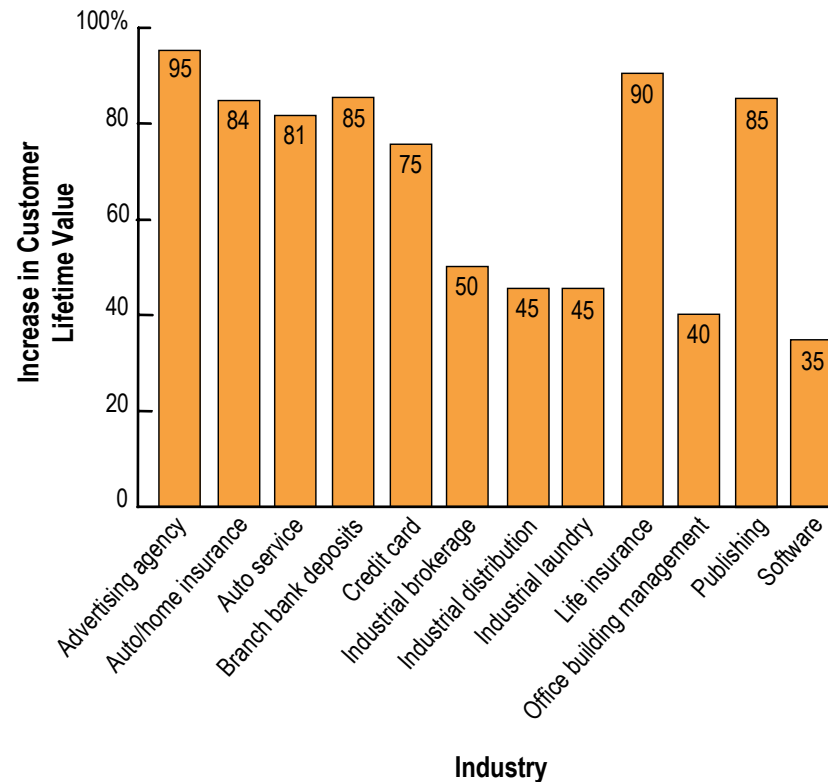
Customer Retention

Small changes in customer retention drive major changes in sales and market share



Customer Retention

Impact on CLV: – *Increase customer retention rate by 5 percent*



Customer Retention

Margin Multiple Values

Retention Rate (r)	Discount Rate (d)			
	8%	12%	16%	20%
60%	1.25	1.15	1.07	1.00
70%	1.84	1.67	1.52	1.40
80%	2.86	2.50	2.22	2.00
90%	5.00	4.09	3.46	3.00
95%	7.31	5.59	4.52	3.80

All things equal, at a 12% discount rate, increasing retention rate from 60% to 95% improves the margin multiple by a factor of five.

Customer Retention

Margin Multiple Values: *Reduce Discount Rate*

Retention Rate (r)	Discount Rate (d)			
	8%	12%	16%	20%
60%	1.25	1.15	1.07	1.00
70%	1.84	1.67	1.52	1.40
80%	2.86	2.50	2.22	2.00
90%	5.00	4.09	3.46	3.00
95%	7.31	5.59	4.52	3.80

All things equal, at a 90% retention rate, decreasing discount rate from 20% to 8% improves the margin multiple by a factor of less than two.

Customer Retention

Margin Multiple Values: *Increase Customer Retention Rate*

Discount Rate (<i>d</i>)					
Retention Rate (<i>r</i>)	8%	12%	16%	20%	
60%	1.25	1.15	1.07	1.00	
70%	1.84	1.67	1.52	1.40	
80%	2.86	2.50	2.22	2.00	
90%	5.00	4.09	3.46	3.00	
95%	7.31	5.59	4.52	3.80	

Customer Retention

Impact on Firm Growth

- Consider two companies:

	Company 1	Company 2
Retention rate (r)	95%	90%
Acquisition rate (a)	15%	15%

- How many years will it take each company to double the size of its customer base?

7 years

14 years

Customer Retention

Long-Run Market Share Scenarios

Scenario A			Scenario B			Scenario C		
Year 2			Year 2			Year 2		
Year 1	Ram	Rani	Year 1	Ram	Rani	Year 1	Ram	Rani
	Ram	Rani		Ram	Rani		Ram	Rani
	80%	20%		90%	10%		95%	5%
	20%	80%		20%	80%		20%	80%

Long-run market shares

Ram – 50%	Ram – 67%	Ram – 80%
Rani – 50%	Rani – 33%	Rani – 20%

Address Current/Potential Customers

Current Customers

$$\text{CLV} = \frac{m \times r}{(1 + d - r)}$$

Potential Customers

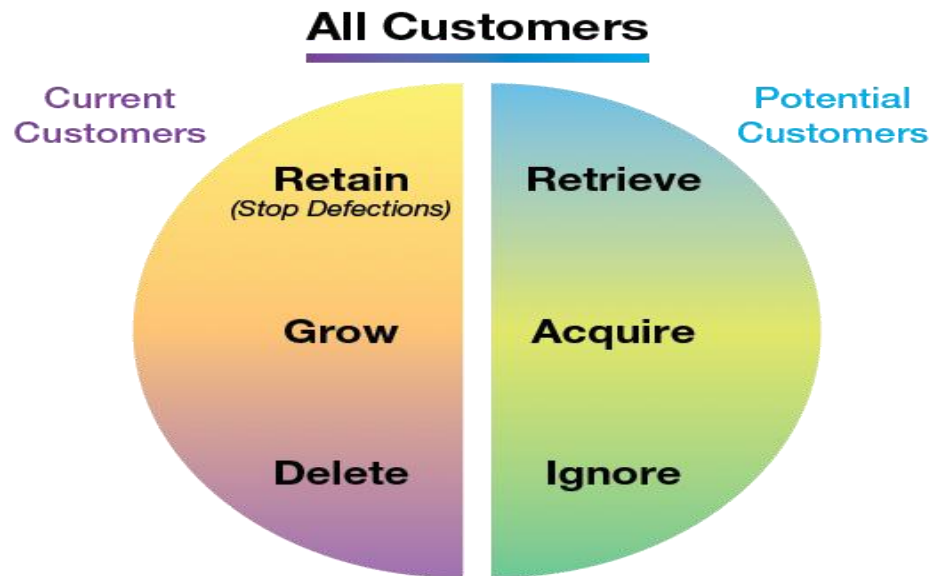
$$\text{CLV} = \frac{m \times r}{(1 + d - r)} - \text{AC}$$

Decision Rule: Acquire potential customers when expected CLV is greater than the acquisition cost.

Address Current/Potential Customers

- Some customers bring value to the firm; some do not
- Measuring customer profitability is critical
- A critical firm challenges is to identify the right customers

Options



Address Current/Potential Customers

How to Acquire Customers

- **Independent marketing activities**
- **Affiliations**
- **Channel strategies**
- **Firm and business acquisitions**

Customer Revenue and Profit Rules

80:20:120 Rule

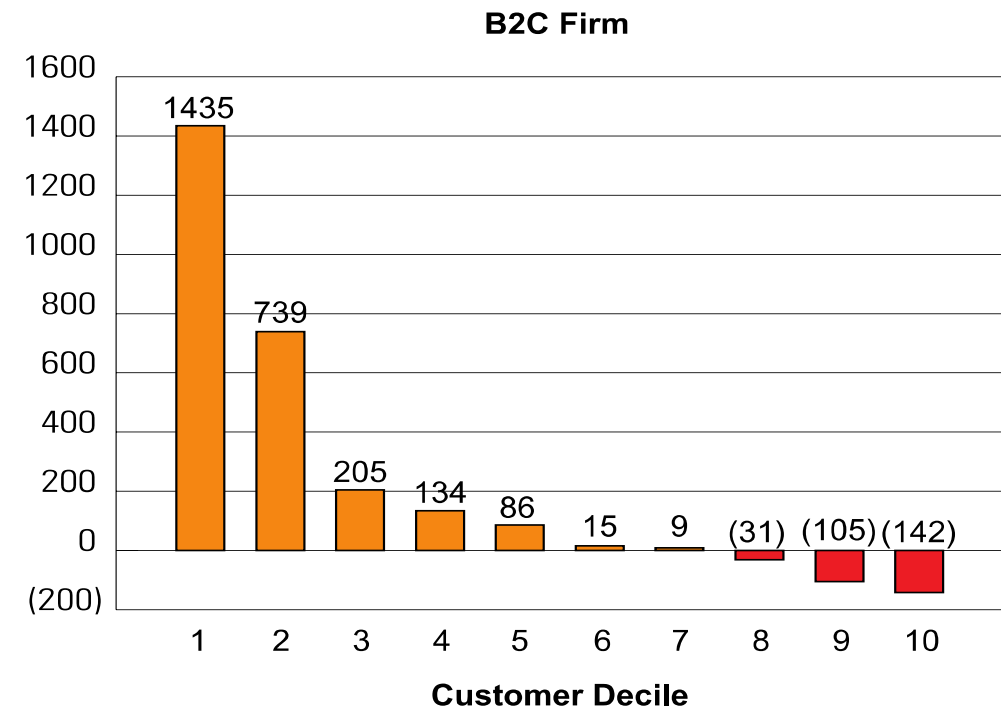
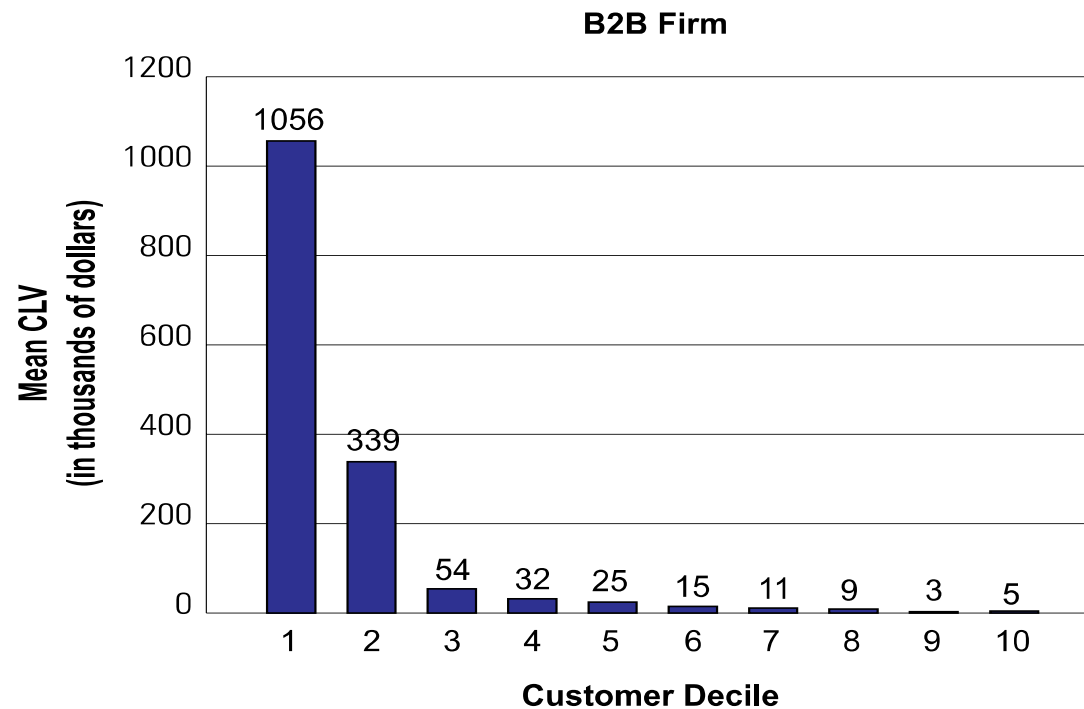
80 percent of revenues
20 percent of customers
120 percent of profits

20:80(20) Rule

20 percent of revenues
80 percent of customers
Reduce profits by 20 percent

Customer Revenue and Profit Rules

CLV Distribution by Customer



Source: V. Kumar and D. Shah, "Expanding the Role of Marketing: From Customer Equity to Market Capitalization," *Journal of Marketing*, 73 (November 2009), pp. 119-136.

Customer Revenue and Profit Rules

Unprofitable/Low-Profit Customers: *Options*

Selling

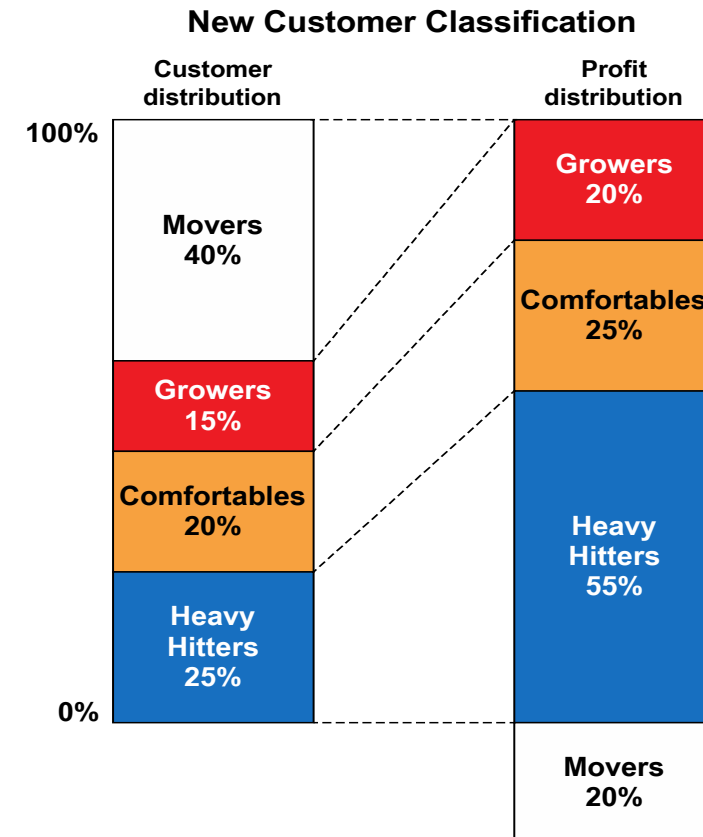
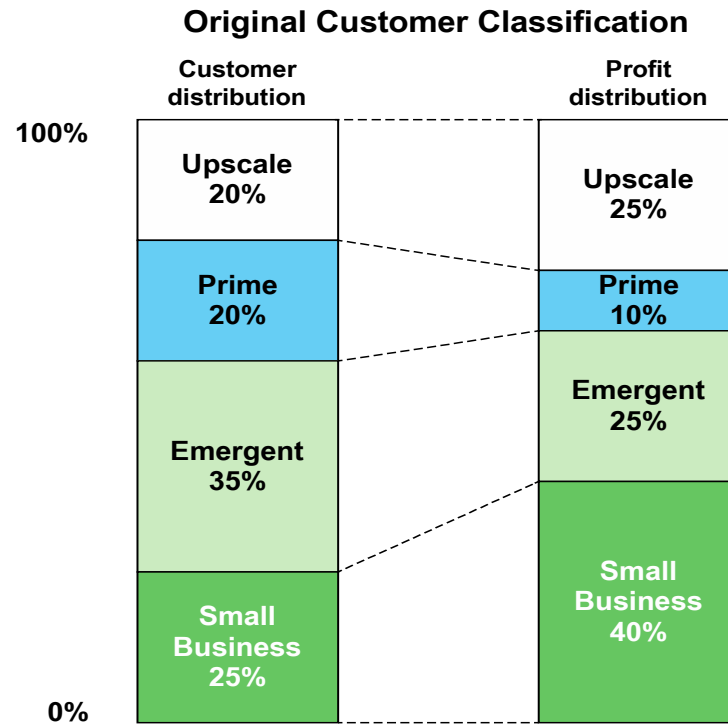
- Part-time salesforce
- Shift to telesales websites
- Assign to agents/distributors
- Outsource selling effort
- Alliances

Non-Selling

- Traditional direct marketing
- Electronic contact – e-mail
- Raise prices
- Stop serving

Customer Revenue and Profit Rules

Current Customers: *Illustration – Financial Services Firm*



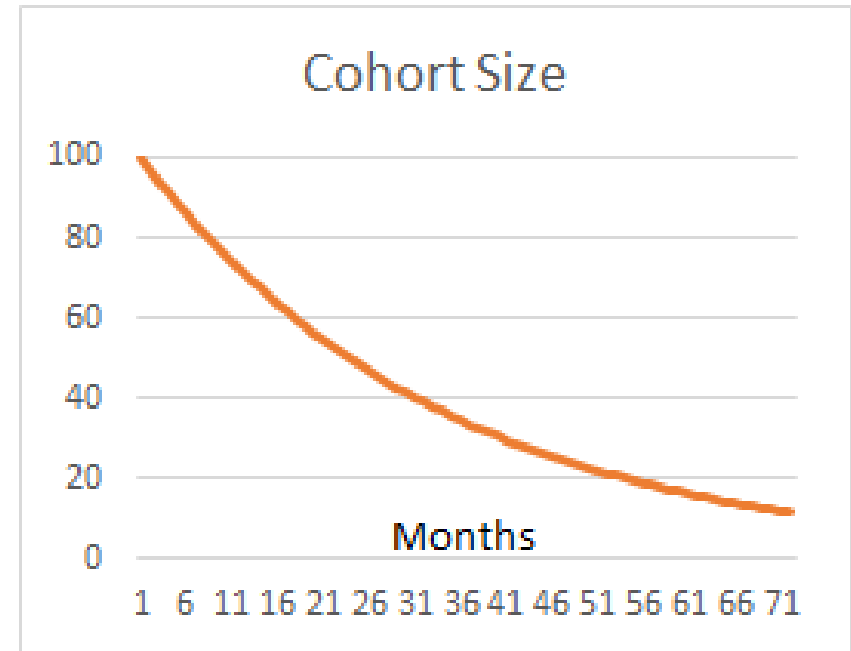
CAC – Cost to Acquire a Customer

- CAC is defined as follows:

$$CAC = \frac{\text{Sum of all Sales \& Marketing expenses}}{\text{No of new Customers Added}}$$

Customer Lifetime

If you start with a cohort of 100 customers and apply a constant churn rate 3% every month, you'll get an exponential decay:



$$\text{Customer Lifetime} = \frac{1}{\text{Customer Churn Rate}}$$

- If the Monthly customer churn rate is 3%, then the Customer Lifetime will be $1/0.03$ which is 33 months.
- If the Annual customer churn rate is 20%, then the Customer Lifetime will be $1/0.20$ which is 5 years.

Metrics for Churn

# of new Customers	The number of new customers added this month
# of churned Customers	The number of customers lost due to churn this month
Net New Customers	Net New Customers = # of new Customers – # of churned Customers This is the net number of new customers added once lost customers due to churn has been taken into account.
% Customer Churn	$\% \text{ Customer Churn} = \frac{\# \text{ of customers who churned}}{\text{Total \# of customers}}$
% MRR Churn	Defined as lost revenue due to churned customers as a percentage of total recurring revenue. $\% \text{ MRR Churn} = \frac{\text{Churned MRR}}{\text{Previous Months MRR}}$ (See below for a description of why this is different to % Customer Churn.)
% MRR Expansion	Defined as the expansion revenue from existing customers as a percentage of total revenue. $\% \text{ MRR Expansion} = \frac{\text{Expansion MRR}}{\text{Previous Months MRR}}$
% Net MRR Churn	$\% \text{ Net MRR Churn} = \frac{\text{Churned MRR} - \text{Expansion MRR}}{\text{Previous Months MRR}}$ This is the number that will go negative if the Expansion revenue from existing customers starts to outstrip the lost revenue from churn. Getting to negative Net MRR Churn is a great goal for a SaaS company.

Lifetime Value of Customer

Average Revenue Per Customer

$$LTV = ARPC \times Customer Lifetime$$

Average Revenue Per Account (ARPA)

$$LTV = \frac{ARPA}{Customer Churn Rate}$$

What is a Repeatable, Scalable Sales Model?

- The process that you go through to acquire a paying customer is clearly repeatable.
 - If your process involves salespeople, you can add new hires and they can achieve the same productivity level as the original sales team.
 - If it is a touchless web sales model, your web traffic converts in a predictable way through your web site.
- The process is scalable.
 - You can increase the sources of your leads and/or web traffic without reaching a near-term limit.
 - The resources (e.g. salespeople) in your conversion funnel can easily be scaled without reaching a near-term limit.
- Your cost to acquire a customer [<http://www.forentrepreneurs.com/startup-killer/>] (CAC) is significantly less than the amount you can monetize them over the customer's lifetime.
 - In a SaaS business It is recommend that LTV should be more than three times higher than CAC.
 - It should also be possible to recover CAC in less than 12 months for a capital-efficient startup.
 - Lifetime value (LTV) should be calculated using gross profit (not revenue) after cost of goods, cost to serve and cost of on-boarding.

Lesson Learned

- Look for ways to get customers to pay in advance
 - Depending on the cost of your capital, this can be worth fairly large discounts

Calculate Churn

- In a quarter, a start-up internet provider acquired 1000 new subscribers. In that same quarter, the company also lost 120 subscribers. The churn rate would be:

Customer Lifetime Value

Objective

- What is CLV
- Why CLV gained importance

Introduction

- Customers are the key asset of any business; without customers there is no business.

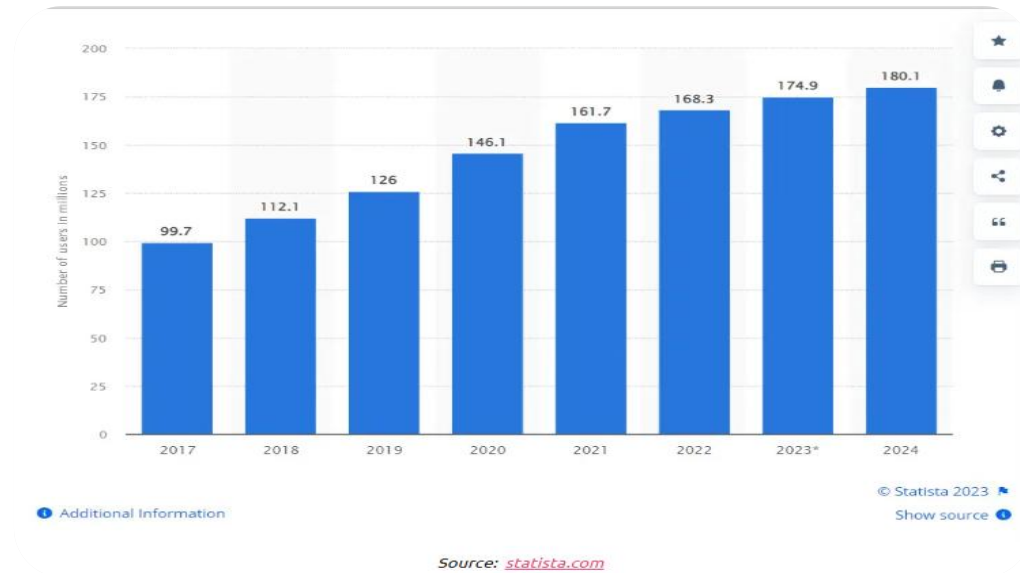
What is CLV

- Customer lifetime value is a measure of customer profitability over time
- CLV can be defined as *“a measure of a customer’s aggregate profit to the firm over the total time that the customer deals with the firm”*
- CLV is calculated as a single dollar number, which summarizes the net profit/loss position of the customer’s total relationship with the firm
- It is calculated on per customer basis, but is more usually determined for an the average customer within a particular market segment
 - A firm will calculate multiple CLV’s for different customer segments

Amazon

- **Price:** Around July 31, 2025, closing prices were near \$234, with daily movements, according to [StatMuse](#).
- **Market Cap:** By June 30, 2025, Amazon's market cap was around \$2.371 trillion

Amazon Prime



Amazon Prime Customer Retention



69% of Members convert from Trial Membership to Paid Membership



93% of Members Continue to Renew Membership after 1 Year



98% of Members have been Prime Members for 2 years

Influencer MarketingHub



98% of Members have been Prime Members for 5 years

MarketingHub

CLV

- “Customer lifetime value is the dollar value of a customer relationship based on the present value of the projected future cash flows from the relationship.”

Marketing Metrics: the definitive guide to measuring marketing performance (Farris et al 2010)

- CLV takes both backward looking and forward looking
 - Calculate value of past customers
 - Use that information to project forward

Use of CLV

- Examine how much to spend to acquire a customer.
- Examine how aggressively to spend to retain a particular customer or group of customers

Future Value of Customer

Expected Customer Lifetime in Months	20
Average Gross Margin per Month per Customer	\$100
Average Marketing Costs per Month per Customer	0
Average Net Margin per Month per Customer	\$100
Customer Lifetime Value	\$2000

NPV or Discount Rate

Discount rate converts future cash flows (that is revenue/profits) into today's money for the firm.

- \$100 into a bank account today
- 10% interest,
- then in 12 months' time you would have \$110 in the bank.
- **\$110 next year is equivalent to \$100 today.**
- Net Present Value (NPV)= to evaluate investments
- CLV =to evaluate customer relationships
- CLV is the expected NPV of the cash flows from a customer relationship

CLV is defined as the discounted sum of all future customer revenue streams minus product and servicing costs and remarketing costs

Calculating CLV

- The Base CLV Model

- Gross Margin per Subscription Customer = \$100
- Retention Expense per customer - \$50
- Contribution Margin per customer = $M - R = \$50$
- Retention rate = $r = 80\%$
- Number of customers who joined Subscription in July 2025= 100

Month	Number of Customer	Total Net profit	Total Net profit	Present Value of Net profit
July 2015	100	$[M-R]*100$		
August 2025	$r * 100 = 80$	$r * 100 * [M - R]$		
Sept 2025	$r * (r * 100) =$	$r^2 * 100 * [M - R]$		
Oct 2025	$r * (r * (r * 100))) =$	$r^3 * 100 * [M - R]$		

The Base CLV Model

\$M	Contribution per period from active customers. Contribution = Sales Price – Variable Costs
\$R	Retention Spending per period per active customer.
r	retention rate (fraction of current customers retained each period)
d	discount rate per period
Present Value of net profit calculation is extended up to infinity...	

$$\text{CLV} = [\$M - \$R] \times [(1 + d) / (1 + d - r)]$$

The Base CLV Model

$$\text{CLV} = [\$M - \$R] \times [(1 + d) / (1 + d - r)]$$



**Short-Term
Margin**



**Long-Term
Multiplier**

The Base CLV Model

$$\text{CLV} = [\$M - \$R] \times [(1 + d) / (1 + d - r)]$$



**Short-Term
Margin**



**Long-Term
Multiplier**

- Examine for organization if retention increases
 - How does the discount rate change?
 - How does long term multiplier change?
 - How does CLV change?

Applying CLV

Amazon charges \$20.95 per month. Variable costs are about \$2.50 per account per month. With marketing spending of \$6 per year, their attrition is only 0.5% per month. At a monthly discount rate of 1%, what is the CLV of a customer?

Recalculate

- If Amazon cuts retention spending from \$6 to \$3 per year, they expect attrition will go up to 1% per month
 - **Should they do it?**

Calculate

- **Fitness Gym is trying to figure out how much each of their customers is worth. What is the CLV for an average club membership?**
 - Monthly membership dues: \$160 per person
 - Variable costs for the membership and retention: \$30 per member
 - Length of the average active membership: 30 months

CLV Extension

CLV - Time Horizon

Percent of CLV Accruing in First Five Years						
Discount Rate	Retention Rate					
	40%	50%	60%	70%	80%	90%
2%	99	97	93	85	70	47
4%	99	97	94	86	73	51
6%	99	98	94	87	76	56
8%	99	98	95	89	78	60
..						
...						
20%	100	99	97	93	87	76

Source: Farris, Paul, Neil Bendle, Phillip Pfeifer, and David Reibstein (2010), *Marketing Metrics*, 2nd Edition, FT Press.

CLV – Initial Margin

- **Customer pays before using the service**

- Subscription services

- **Customer pays after using the service**

- credit cards

$$CLV = [\$M - \$R] \times [(1+d)/(1+d-r)]$$

$$CLV = [\$M - \$R] \times [(1+d)/(1+d-r)]$$

CLV – Cohort

- Cohort = customers acquired at the same time period (month, quarter or year)
- Since retention changes with time since acquisition, CLV calculations are better if they are done separately for each cohort

Calculate

- You are trying to maximize CLV based on the following information. How much should you spend on marketing and retention (R) to maximize CLV?
 - $M = \$50$
 - $d = .01$
 - If $R = \$49$, then $r = 99\%$
 - If $R = \$40$, then $r = 90\%$
 - If $R = \$30$, then $r = 75\%$
 - If $R = \$20$, then $r = 60\%$
 - If $R = \$10$, then $r = 50\%$
- A. \$40
- B. \$30
- C. \$10
- D. \$49

Calculate

- **Consider this subscription service. What is the CLV for an average customer?**
 - Billing: End of the month
 - Monthly margin: \$100 per customer
 - Annual retention spending: \$780 per customer
 - Monthly retention (r): 97%
 - Discount rate (d): 3%

A. \$16.17

B. \$17.17

C. \$600.83

D. \$565.95

CLV – Contractual vs Non Contractual

- Amazon and Netflix have a contract with customers
 - They know when a customer unsubscribes to the service
 - This helps in knowing lifetime duration and retention rate
- What if a customer does not sign a contract to use a service?
 - e.g., grocery stores like JioMart, DMart

Customer Lifetime Value



Customer Lifetime Value

- It provides a forward-looking measure of the customer relationship
- It connects marketing strategies to financial consequences
- Strategic marketing actions can be evaluated based on whether they improve customer retention and lifetime value

Further Readings

- David Skok- <https://www.forentrepreneurs.com>
- Geoff Fripp - <https://www.clv-calculator.com/>